

TrES-3b

R-2786

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150706 · created 2026-09-04T00:33:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457209.7317 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.056
Transit depth (Rp/Rs)^2	2.51 +/- 1.79 [%]
Orbital Inclination (inc)	77.34 +/- 2.87
Airmass coefficient 1 (a1)	0.35 +/- 0.21
Airmass coefficient 2 (a2)	1.1 +/- 1.38
Scatter in the residuals of the lightcurve fit is	54.96 %
Best Comparison Star	#1 - [145, 137]
Optimal Aperture	0.87
Optimal Annulus	6.01
Transit Duration (day)	0.011 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.056 vs 0.16309 (deviation 0.08σ)
Duration fit vs published	0.011 d vs 0.0567 d (deviation 2.07σ)
Mid-time O–C	-1.7 min
Depth SNR	2.6
Residual scatter	54.96 %

Light curve

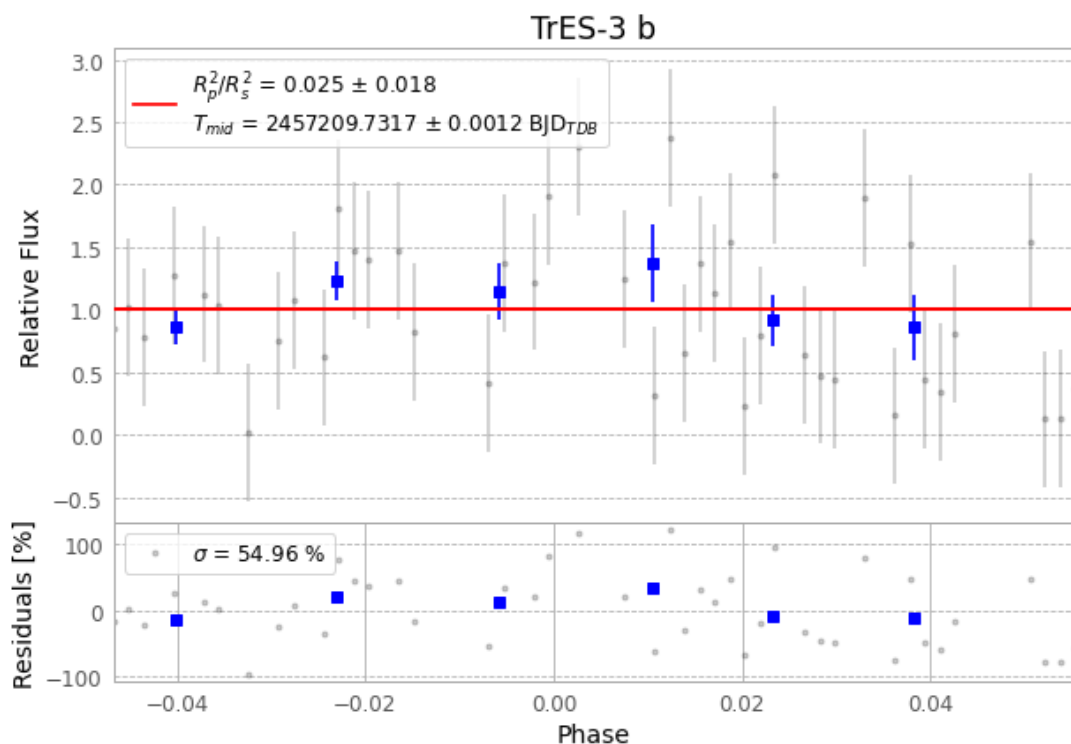


Figure 1 — FinalLightCurve_TrES-3 b_2015-07-05.png (SHA-256 b9f16ea1b823bba7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [325, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fae3e76658457a27...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3150706035712.FITS → TRES-3150706071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150706.log (d9e2611c2104...), FinalLightCurve_TrES-3 b_2015-07-05.png (b9f16ea1b823...), FinalLightCurve_TrES-3 b_2015-07-05.csv (fdf516f4fe86...)

Compute resources

Wall time 181.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 00:33Z.